

John Heck

Norwich, CT | (860) 705-4802

John.Heck@uconn.edu | linkedin.com/in/john-heck/

US DoD Secret Clearance (Current)

Education

University of Connecticut

Bachelor of Science in Mechanical Engineering, Class of 2028

Concentration: Aerospace

GPA: 3.64/4.0

Storrs, CT

Aug. 2024 - Present

Work Experience

General Dynamics Electric Boat

Groton, CT

Propulsion Plant and Fluid Systems - Design & Engineering Intern

Jun. 2026 - Aug. 2026

- Resolved hardware thread interference by redesigning adapter geometry with GD&T and specifying new O-ring parameters, preventing leakage and enabling critical testing.
- Dispositioned fluid-pipe routing nonconformances by evaluating geometric deviations and verifying zero impact to vessel acoustic signature, avoiding part rejection.
- Identified a legacy O-ring thermal-rating mismatch with MIL/AS586 specs; issued a conditional Use-As-Is disposition requiring downstream replacement for NAVSEA compliance.
- Supported manufacturing transition of an insourced fluid component by observing first-run fabrication and verifying production readiness.
- Audited 40+ manufacturing Excel sheets against Teamcenter baselines and Engineering Change Notices to verify configuration compliance.
- Corrected gage-valve orientation and updated schematics to ensure NAVSEA compliance across future COLUMBIA-class hulls.

Clubs and Organizations

UConn Electric Boat Racing Team - Fluid Dynamics Team Lead

Jan. 2026 - Present

- Directing a 3-person team to model propeller fluid interactions, establishing an analytical pipeline to investigate localized pressure drops, cavitation risk, and resulting loss of hydrodynamic thrust.
- Leading weekly technical design reviews to define boundary conditions, track fluid dynamics milestones, and align the sub-team on upcoming hydrodynamic testing objectives.
- Modeling the SEADOG II hull and propeller geometry in SolidWorks to establish boundaries for fluid dynamics testing.

UConn Fencing - Vice President

May 2026 – Present

- Managed operational logistics and budget allocations for a competitive collegiate team, utilizing platforms such as AskFRED and FencingHub to optimize tournament prep and registration for USA Fencing events.

Academic Projects

RC Car Teardown – Junior Design

Aug. 2026 – Present

- Executed a hardware teardown to establish kinematic baselines.
- Engineering a lightweight chassis integration for upgraded battery architecture, reducing total vehicle mass to optimize the power-to-weight ratio and acceleration.

Skills

Technical: SolidWorks, Python, MATLAB, ANSYS (Fluent, Workbench, Mechanical) Teamcenter, Git/GitHub, Excel

Relevant Coursework: Intro to CFD, Intro to FEA, Dynamics, CAD and Manufacturing, Fluid Dynamics, Thermodynamics, Data Structures and Algorithms, Mechanics of Materials